George Marselis | george.marselis@vetinst.no | NVI 2026

IRIDA REST API Reference

irida.vigasp.vetinst.no:8080/irida-23.01.3/api

Authentication

POST	/api/oauth/token	Get bearer token. Params: client_id, client_secret, grant_type=password, username, password. Token valid for configured duration (default 12h). Include in all requests as: Authorization: Bearer {token}

Projects

GET	/api/projects	List all projects the authenticated user has access to
GET	/api/projects/{id}	Get a single project by ID
GET	/api/projects/{id}/samples	List all samples in a project
GET	/api/projects/{id}/samples/metadata	Get metadata for all samples in a project
GET	/api/projects/{id}/users	List all users with access to a project
GET	/api/projects/{id}/analyses	List analyses shared with a project
GET	/api/projects/{id}/hash	Get project hash - integer checksum for change detection. Store previous hash to detect changes; value alone is not useful without a baseline.
POST	/api/projects	Create a new project. Required: name (min 5 chars). Optional: projectDescription, organism
POST	/api/projects/{id}/samples	Create a new sample in a project. Required: sampleName (min 3 chars). Optional: organism, description, collectionDate, collectedBy
POST	/api/projects/{id}/users	Add a user to a project. Required: userId (username). Optional: role (PROJECT_USER | PROJECT_OWNER)
PATCH	/api/projects/{id}	Update project properties. Fields: name, projectDescription, organism
DELETE	/api/projects/{id}/users/{username}	Remove a user from a project

Samples

GET	/api/samples/{id}	Get full metadata for a single sample
GET	/api/samples/{id}/sequenceFiles	List all sequence files for a sample (single-end + paired-end). Does not surface pair identifier - call /pairs separately to get pair_id for downloads.
GET	/api/samples/{id}/sequenceFiles/pairs	List paired-end sequence files only
GET	/api/samples/{id}/sequenceFiles/unpaired	■ BUG (IRIDA 23.01.3): returns HTTP 500. Servlet parses "unpaired" as Long fileId (NumberFormatException). Workaround: filter /sequenceFiles excluding IDs in /pairs. See issue #4.
GET	/api/samples/{id}/sequenceFiles/fast5	List FAST5 files for a sample
GET	/api/samples/{id}/metadata	Get unstructured key-value metadata for a sample (separate from structured fields at /api/samples/{id}). See issue #7.
GET	/api/samples/{id}/assemblies	List assemblies for a sample (populated only by IRIDA pipeline runs)
PATCH	/api/samples/{id}	Update sample metadata: sampleName, description, organism, collectionDate, collectedBy, strain, isolate, latitude, longitude, geographicLocationName, isolationSource

Sequence Files

GET	/api/samples/{id}/pairs/{pairId}/files/{fileId}	Get or download an individual sequence file. Requires Accept: application/fastq to download. fileName carries R1/R2 convention. NOTE: ONLY endpoint where Accept header has effect - all others return text/plain regardless.
POST	/api/samples/{id}/sequenceFiles	Upload a single-end FASTQ file. Multipart form field: file=@/path/to/file.fastq.gz
POST	/api/samples/{id}/sequenceFiles/pairs	Upload a paired-end FASTQ pair. Multipart form fields: file1=@R1.fastq.gz file2=@R2.fastq.gz
POST	/api/samples/{id}/sequenceFiles/fast5	Upload a FAST5 file. Multipart form field: file=@/path/to/file.fast5
DELETE	/api/samples/{id}/sequenceFiles/{fileId}	Delete a sequence file from a sample

Sequencing Runs (admin only)

GET	/api/sequencingrun	List all sequencing runs. Returns empty resources array for non-admin tokens.
GET	/api/sequencingrun/{id}	■ BUG (IRIDA 23.01.3): HATEOAS link points to /api/sequencingrun/miseqrun which throws HTTP 500 - same routing bug as /unpaired (NumberFormatException). See issue #6.
POST	/api/sequencingrun	Create a sequencing run. Required: layoutType (SINGLE_END | PAIRED_END), sequencerType (miseq | nextseq | miniseq | directory). uploadStatus auto-set to UPLOADING.
PATCH	/api/sequencingrun/{id}	Update sequencing run uploadStatus. Values: UPLOADING | COMPLETE | ERROR

Analysis Submissions

GET	/api/analysisSubmissions	List all analysis submissions for the current user
GET	/api/analysisSubmissions/analysisType/phylogenomics	List phylogenomics analyses. ■ SLOW: queries Galaxy synchronously per submission - can saturate all uwsgi workers making IRIDA unresponsive. See issue #8.
GET	/api/analysisSubmissions/analysisType/assembly	List assembly and annotation analyses. ■ SLOW: same Galaxy-blocking behaviour. Expect 15-60+ seconds.
GET	/api/analysisSubmissions/analysisType/sistr	List SISTR typing analyses. ■ SLOW: same Galaxy-blocking behaviour.
GET	/api/analysisSubmissions/{id}	Get a single analysis submission. NOTE: analysisType field is NOT present here - only available via /{id}/analysis when state=COMPLETED.
GET	/api/analysisSubmissions/{id}/analysis	Get completed analysis outputs (only when state=COMPLETED). Contains analysisType and HATEOAS links to output files. IRIDA ignores Accept header - all output files returned as text/plain.

Users (documented as admin-only - confirmed accessible with unprivileged token, see issue #5)

GET	/api/users	List all users in the system
GET	/api/users/{username}	Get a single user by username
GET	/api/users/{username}/projects	List projects the user has access to

Miscellaneous

GET	/api	Root endpoint - returns links to all top-level resources (HATEOAS)
GET	/api/version	Get IRIDA server version string. Requires Bearer token.

All endpoints require: Authorization: Bearer {IRIDA_TOKEN} | Responses are JSON (Accept: application/json) | Timestamps are milliseconds since epoch | 403 = not permitted | 401 = no token | 405 = verb not supported